Homework 3

Due March 4, 2026 at 9 pm

Problem 1.

It is not well understood how changes in the genetic makeup of individuals may alter the behavior of groups of animals. Zebrafish are known to display a variety of group behaviors, such as swimming together in an aligned group. Florian's experiment was conducted in order to investigate whether mutations in a particular gene known to affect social behavior in humans (*scn1lab*, which stands for *sodium channel, voltage-gated, type I like, alpha b*) will affect alignment in larval zebrafish that are 7 days post-fertilization (i.e., 7 dpf).

In this experiment, i represents an index for a group of 5 larval zebrafish that are 7 dpf. The notation G_i^{obs} is used to denote the genotype of the group; specifically, $G_i^{obs} = 0$ for a wildtype group and $G_i^{obs} = 1$ for a *scn1lab* mutant group. The ith group outcome is denoted with Y_i^{obs}.

These data were reported in *Collective behavior emerges from genetically controlled simple behavioral motifs in zebrafish* (Harpaz et al., 2021).

Take a look at Panel A of Figure 2 from Florian's paper (Harpaz et al., 2021), particularly the third plot, as well as Figure 2's legend.

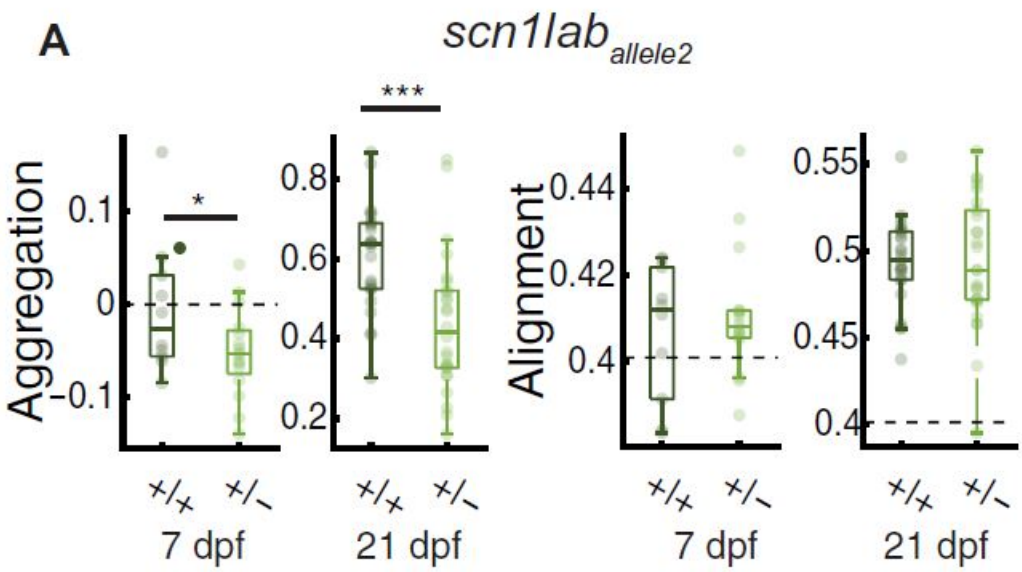

Fig. 2. Single-gene mutations affect aggregation and alignment of developing zebrafish.

+/+, +/−, and −/− refer to sibling-controlled wild-type and mutant fish. (**A**) Left: At 7 dpf, *scn1lab*$^{+/-}$ fish are more dispersed than wild-type siblings ($P_{\text{Fisher}} \approx 0.036$, $N_{+/+} = 10$, $N_{+/-} = 16$ groups, Cohen's $d = -0.8$). Dashed lines represent values of shuffled groups. At 21 dpf, fish are more aggregated. *Scn1lab*$^{+/-}$ aggregate less than *scn1lab*$^{+/+}$ ($P_{\text{Fisher}} \approx 0.0001$; $N_{+/+} = 25$, $N_{+/-} = 26$ groups, Cohen's $d = -1.09$). Right: Group alignment increases with age; however, we could not detect an effect of the *scn1lab* mutation ($P_{7\ \text{Fisher}} = 0.26$, $P_{21\ \text{Fisher}} = 0.47$). *$P < 0.05$, ***$P < 0.0005$. (**B**) Left: At 7 dpf, *disc1*$^{-/-}$ are less dispersed than wild-type siblings ($P_{\text{Fisher}} \approx 0.0128$; $N_{+/+} = 7$, $N_{-/-} = 17$ groups, Cohen's $d = 1.04$). At 21 dpf, *disc1*$^{-/-}$ show more aggregation compared to wild-type siblings ($P_{\text{Fisher}} \approx 0.0129$; $N_{+/+} = 12$, $N_{-/-} = 13$ groups, Cohen's $d = 0.95$). Right: We could not detect effect of the *disc1* mutation on alignment ($P_{7\ \text{Fisher}} = 0.33$, $P_{21\ \text{Fisher}} = 0.40$). (**C**) Pearson's correlation of alignment and aggregation in *scn1lab*$_{allele2}$ 7-dpf (top) and 21-dpf (bottom) fish. Positive correlation for 21-dpf *scn1lab*$_{allele2}^{+/-}$ fish. Uncertainty regions are the pointwise 95% CI of the linear regression model. (**D**) We could not detect correlation for *disc1*. (**E**) Left: *scn1lab*$_{allele2}^{+/-}$ turn away more from visual clutter at 7 dpf (top) and turn toward clutter less at 21 dpf (bottom). Right: Integral of the curves symmetrized. Repulsion is negative. Attraction is positive. (**F**) Same as in (E) but for *disc1*. Mutants show a flattening of the 7-dpf repulsion curve (top) and an enhancement in 21-dpf attraction (bottom). Bold lines in (E) and (F) represent turning probability calculated from left/right turns of all fish in 5° bins; uncertainty regions are the pointwise 95% CI of a fitted binomial distribution to the events in each bin.

Part of the method section is presented here.

We examined whether single-gene mutations in *scn1lab* and *disc1* have an effect on group aggregation and alignment phenotypes measured at 7 dpf and at 21 dpf (Fig. 2). We used groups of five zebrafish of the same mutant type, which we denote by j. Let G_j denote the mutation status of group j, where $G_j = 0$ for a wild-type group and $G_j = 1$ for a mutant group, and assume a completely randomized assignment of the group mutation status. For example, $G_j = 0$ for a group of five *scn1lab*$_{\text{allele2}}^{+/+}$ fish and $G_j = 1$ for a group of five *scn1lab*$_{\text{allele2}}^{+/-}$ fish. For each group j, we measure an outcome Y_j, such as group aggregation or group alignment. We denote $Y_j(G_j = 0)$ the potential outcome had group j been wild type and $Y_j(G_j = 1)$ the potential outcome had group j been mutant. We test the sharp null hypothesis stating that, for each group j, the potential outcomes are equal, that is, $Y_j(G_j = 0) = Y_j(G_j = 1)$. We use T_{Welch} as our test statistic.

The highlighted $P_{7Fisher}$ is the resulting Fisher-exact p-value that Florian obtained after testing the sharp null hypothesis stating that the *scn1lab* mutation has no effect of alignment of 7 dpf zebrafish. What is the total number of allocations in that controlled experiment? Assume a completely randomized design. Replicate Florian's result (i.e., $P_{7Fisher} \approx 0.26$) using 100,000 randomized allocations and document your answer. Here, assume we define more extreme as "greater than".

In 2019, Wasserstein et al. argued that we should "not base conclusions solely on whether an association or effect was found to be 'statistically significant' (i.e., the p-value passed some arbitrary threshold such as p<0.05)". So, also report a 95% CI for the average causal effect τ on the additive scale.

Describe all your findings.

Problem 2

Consider the simple case with one treated and one control unit in a completely randomized experiment. The estimand, the sample average treatment effect, in this case, is:

$$\tau = \frac{1}{2}\{[Y_1(W_1 = 1) - Y_1(W_1 = 0)] + [Y_2(W_2 = 1) - Y_2(W_2 = 0)]\}.$$

(a) Write the unbiased estimator of τ proposed by Neyman as a function of Y_1^{obs} and Y_2^{obs}.

(b) Write the unbiased estimator of τ proposed by Neyman as a function of the potential outcomes.

(c) To simplify the following calculations of the sampling variance of this estimator, let us define the binary variable $D = 2W_1 - 1$, so D $\in$ {-1,1}. Calculate $\mathbf{E}_W[D]$ and $\mathbf{V}_W[D]$.

(d) Show that the unbiased estimator of τ proposed by Neyman can be written as:

$$\hat{\tau} = \frac{D+1}{2}[Y_1(1) - Y_2(0)] + \frac{1-D}{2}[Y_2(1) - Y_1(0)]$$

(e) Show that the unbiased estimator of τ proposed by Neyman can be written as:

$$\hat{\tau} = \tau + \frac{D}{2}\{[Y_1(1) + Y_1(0)] - [Y_2(1) + Y_2(0)]\}$$

(f) Show that the unbiased estimator of τ proposed by Neyman is unbiased for τ.

Problem 3

H_{Null} states that for each unit i, $Y_i(W_i = 1) - Y_i(W_i = 0) = 0$. The counternull value is a value of an estimand that is supported by as much evidence in the data as the null value. $H_{Counternull}$ states that for each unit i, $Y_i(W_i=1)-Y_i(W_i=0)=a, where $a \neq 0$ is a counternull value (i.e., $H_{Counternull}$ assumes constant and additive unit-level effects). The counternull randomization distribution $\{T^{(r)}_{H_{Counternull}}\}$ is the set of N_{rand} possible values of the test statistic such that:

$$\frac{1}{N_{rand}} \sum_{r=1}^{N_{rand}} \delta(T^{(r)}_{H_{Null}} \geq T^{obs}) = \frac{1}{N_{rand}} \sum_{r=1}^{N_{rand}} \delta(T^{(r)}_{H_{Counternull}} \leq T^{obs}),$$

where $\delta(x) = 1$ if x is true, 0 otherwise.

The counternull p-value is equal to the Fisher-exact p-value, but is specified by defining more "unusual or more unusual" than T^{obs} using the other tail of the counternull distribution. Because they have the same p-value, the null and counternull values are equally consistent with the observed data.

Calculate the counternull value for the CpG site *cg00000029* contained in completelyrandomized.csv and explain your procedure. Use the average outcome difference between the participants exposed to ozone versus clean air for your test statistic. Define "more extreme" as "greater than".

Problem 4

Air pollution exposure has been shown to be associated with DNA methylation, which is an epigenetic process by which methyl groups are added to the DNA molecule. DNA methylation can change the activity of a DNA segment without changing the sequence. Data from a randomized experiment are provided to examine whether exposure to ozone (vs. clean air) changes DNA methylation measured at CpG sites, which are regions of DNA where a cytosine nucleotide is followed by a guanine nucleotide.

Assume a completely randomized experiment in which N_E units are randomly selected from a population of N units to be exposed for two hours to ozone at 0.3 ppm concentration, with the remaining N_U units assigned to clean air. We denote the active exposure by $W_i = 2$ and the control exposure by $W_i = 0$.

In this experiment, assume that each participant has an exposure assignment probability equal to $\frac{N_E}{N}$. After each exposure, DNA methylation (denoted by Y) was measured in bronchial cells and corresponds to the ratio of methylated cytosines to methylated and unmethylated cytosines (i.e., a continuous outcome with [0,1] support).

Provide a 95% "Fisher" interval (FI) for τ_i for the CpG *cg00000029* that you can find in completelyrandomized.csv.